Project Proposal

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Sharad Hasan
Lecturer
Department of Data Science and Engineering
Gazipur Digital University

Iyolita Islam
Lecturer
Department of Computer Science and Engineering
Bangladesh University Of Professionals

Team Members	ID	Name
01	24524203140	Muntaha Binte Mostafa
02	24524203148	Shafia Akter
03	24524203158	Mahfuz Hossain Antor
04	24524203164	Atia Mahmuda Ahana

GridPulse

AI-Driven Smart Grid Load Balancer & Predictive Anomaly Engine

Problem Statement:

The increasing use of renewable energy sources such as solar and wind creates significant fluctuations in electricity generation. At the same time, consumer electricity demand can change rapidly depending on factors such as weather, time of day and unexpected events.

Traditional monitoring systems mainly focus on the current condition of the grid. They may not provide sufficient predictive information about where demand is likely to increase or where an overload may occur.

This can lead to problems such as:

- Unexpected load spikes
- Transformer or transmission-line overload
- Imbalance between electricity generation and demand
- Difficulty in identifying abnormal grid conditions
- Increased risk of localized power interruptions

Therefore, GridPulse aims to provide a web-based platform that combines real-time grid monitoring with AI-based demand forecasting and rule-based safety validation. The system will use a simulated power grid and generated IoT telemetry data to predict possible problems before they become critical.

Target Users:

The primary target users of GridPulse are:

- **Microgrid Operators :**

They can monitor the condition of different nodes and identify overloaded or abnormal areas of a microgrid.

- **Power Utility Engineers:**

Engineers can use the system to analyze electricity demand, identify potential problems, and evaluate simulated grid conditions.

- **Smart City Infrastructure Managers:**

They can use the dashboard to understand overall energy distribution and observe possible demand-related risks.

- **Educational and Research Users:**

The simulated environment can also be useful for students and researchers who want to experiment with smart-grid monitoring and AI-based prediction without requiring physical electrical infrastructure.

Project Objectives:

The main objectives of GridPulse are:

- To develop an interactive web-based dashboard for real-time monitoring of a simulated power grid.
- To visualize grid topology and energy flow using interactive maps, graphs, and charts.
- To process real-time telemetry data and extract important grid-related features such as load, voltage, and capacity usage.
- To develop an AI-based demand forecasting system to predict future electricity demand and possible load spikes.
- To detect abnormal grid conditions such as sudden load increases, unusual voltage changes, and potential overloads.
- To implement a rule-based safety engine that validates AI predictions according to predefined grid capacity limits.
- To provide a shock scenario simulator for testing events such as heatwaves, sudden demand increases, and renewable energy generation failures.
- To build a safe and feasible simulated environment using a 20-node grid without requiring physical power grid hardware.

Major Features:

1. **Interactive Grid Topology Map:**
The system will provide an interactive visual representation of the simulated grid. Users will be able to see different nodes, substations, connections, and their current status.
Each node can display information such as:

- Current load percentage
- Voltage level
- Node status
- Connection status

Different visual indicators can be used to highlight normal, warning, and critical conditions.

2. Dynamic Energy Flow Visualization:
GridPulse will provide an interactive energy-flow visualization showing how electricity moves from generation sources through grid nodes toward consumers.
A Sankey-style diagram can be used to represent the amount of power flowing through different parts of the grid.

3. Real-Time Telemetry Processing:
The system will receive simulated telemetry data at regular intervals.
The analytical engine will process the incoming data and calculate values such as:

- Current load
- Average load
- Load percentage
- Node capacity usage
- Changes in generation and demand

4. AI Demand Forecasting:
An AI/ML model such as **XGBoost** will analyze historical and processed grid data to predict future electricity demand.For example, the system may identify that a particular node is likely to experience a significant load increase within the next few hours.

5.Anomaly Detection:
The system will identify unusual conditions in the simulated grid.
Examples include:

- Sudden load increase
- Unusual voltage change
- Unexpected reduction in generation
- Node capacity approaching a critical level

The dashboard will display appropriate warnings when such conditions are detected.

6. Safety Rule Enforcer:
The AI prediction will not directly control the grid. A separate rule-based safety layer will validate the prediction or recommendation.
For example, if a transmission connection has a predefined capacity of 100 MW, the system will not allow a simulated routing recommendation that exceeds that capacity.
This provides an additional safety layer between AI predictions and simulated grid actions.

7. Shock Scenario Simulator:
Users will be able to create simulated grid events, such as:

- Heatwave causing increased electricity demand
- Solar plant trip causing reduced generation
- Sudden consumer demand increase
- Failure of a simulated grid node

The system will then show how the simulated grid responds to the event.

8 . Dashboard and Alerts:
A centralized dashboard will present:

- Current grid status
- Load information
- Energy flow
- AI predictions
- Detected anomalies
- Safety warnings
- Simulation results

Proposed Technology Stack:

Frontend

- React / Next.js – For developing the web application and dashboard
- D3.js – For interactive grid topology visualization
- Recharts – For charts and data visualization
- Tailwind CSS – For responsive and modern UI design

Backend

- Python FastAPI – For REST APIs and real-time backend communication
- WebSockets – For transmitting live telemetry and grid updates
- Celery – For background processing tasks
- Redis – For temporary data, task management, and real-time communication

Database

- PostgreSQL – Main relational database
- TimescaleDB – For efficiently handling time-series telemetry data

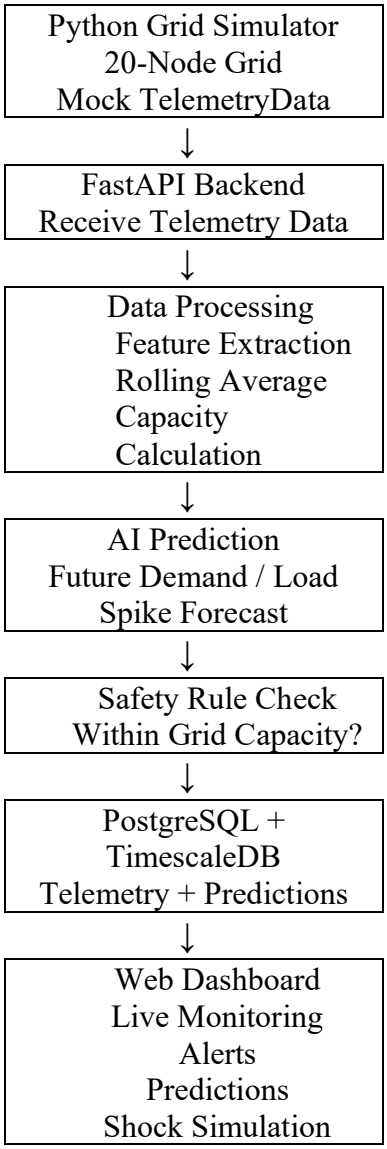
Frameworks

- Pandas – Data processing and analysis
- NumPy – Numerical computation
- Scikit-Learn – Machine learning utilities and preprocessing
- XGBoost – Demand forecasting and predictive modeling

Simulation

- Python-based Grid Simulator – To generate realistic mock IoT telemetry data from the 20-node simulated grid.

Basic Syatem Workflow:



Significance:

The main uniqueness of GridPulse is that it does not depend only on AI prediction. It combines AI-based forecasting with deterministic safety rules.

A machine learning model may predict that a certain power-routing decision could reduce load. However, the recommendation must still satisfy predefined grid constraints before being considered valid.This creates a dual-engine approach: **AI Prediction + Engineering Safety Rules.**

Another important feature is the interactive simulation environment. Instead of requiring expensive physical grid hardware, GridPulse uses a simulated 20-node grid. This makes the project safer, more affordable, and more practical for an academic project.

The project is also significant because it demonstrates how modern web technologies, data processing, visualization, and machine learning can work together in a smart-grid environment.

Feasibility & Project Scope:

Feasibility

GridPulse is feasible within the SDP course duration because the project will use a simulated environment rather than a real electrical grid.

No physical power-generation equipment, transformers, or IoT hardware will be required. A Python-based simulator will generate mock telemetry data and send it to the application.

The use of established technologies such as React/Next.js, FastAPI, PostgreSQL, D3.js, and XGBoost also makes the implementation technically achievable.

Project Scope

To keep the project manageable, the following limitations will be applied:

- The system will use a 20-node simulated grid.
- Real electrical hardware will not be connected.
- IoT data will be generated by a Python simulator.
- AI prediction will focus mainly on electricity demand/load forecasting.
- Power rerouting will be simulated and visualized, not physically performed.
- The system will focus on monitoring, prediction, anomaly detection, safety validation, and simulation.

These limitations ensure that the project remains realistic and achievable within the course timeline.

Expected Outcome:

The expected outcome of GridPulse is a full-stack web application that provides an interactive and intelligent smart-grid monitoring environment.

After completion, the application will allow users to:

- Monitor the simulated grid in real time
- View grid topology and energy flow
- Observe node load and capacity
- Predict future electricity demand using AI
- Detect abnormal grid conditions
- Receive warnings for potential overloads
- Validate recommendations using safety rules
- Simulate different grid shock scenarios
- Observe the effect of simulated events on the grid

Ultimately, GridPulse will demonstrate how AI, real-time data processing, web-based visualization, and engineering safety rules can be combined to create a practical smart-grid management solution.

The final application will serve as a simulation and decision-support platform, rather than a system for controlling a real physical power grid.